

Total no. of Pages: 1

Roll no.

Degree: **B.Tech (MAC)**

Semester: 7

MID-SEMESTER EXAMINATION, September, 2024

Course Title: **Cryptography**

Course Code: **CMCSE50**

Duration: **1:30 Hours**

Max. Marks: 25

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	Explain non-repudiation and its significance as a security service.	2	CO1
1b	Diagrammatically show the launch of a traffic analysis attack. Which type of attack is it and how is it different from Replay attack?	3	CO1
2a	What role does the keyword play in the Vigenère cipher?	2	CO2
2b	How does the size of the key in symmetric encryption affect the security of the algorithm? Discuss the implications of using shorter vs. longer keys.	3	CO1
3a	Explain the working of One-time pad cipher.	2	CO2
3b	Provide the rules used by Play-fair cipher for encryption. Decrypt the ciphertext "WEHFRM GXOBK" using the Playfair cipher with the keyword "CONFIDENTIAL."	3	CO3
4a	On what basis do we define the strength of a cipher? Explain when will you call a cipher computationally secure?	2	CO1
4b	Define Cryptanalytic attacks. Give examples of ciphers that are susceptible to frequency analysis attacks?	3	CO2
5a	Explain the security of DES. Under what type of attacks is this algorithm secure and how?	2	CO3
5b	Diagrammatically provide the flow Feistel Encryption and Decryption.	3	CO3